

Week5

Query Enhancement

From last week

Parent=Semantic, Child = RecursiveTextSplitter

RAM usage for embeddings: 3.3GB

EmbeddingTracker] — Summary

Total tokens : 5,417,362

Total cost : \$0.7043

Model : text-embedding-3-large (\$0.13/1M tokens)

Prompt Caching

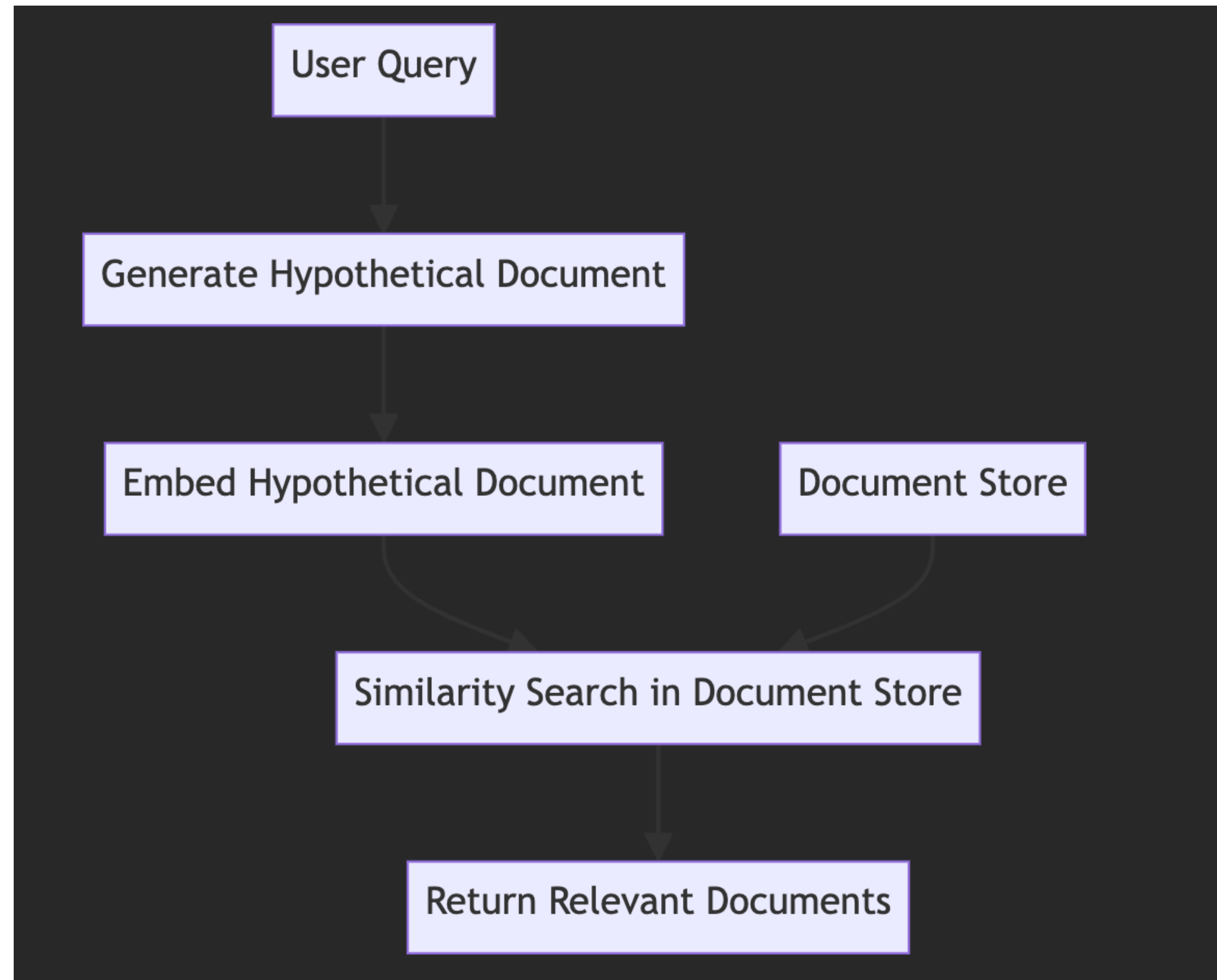
Providers like Anthropic cache

Key Value Vectors (the ones inside the transformer)

Query Transformation

1. Query Rewriting
2. Step-back Prompting
3. Sub-query Decomposition

HyDE



What about the following?

Apply retrieval on knowledge base
before generating a hypothetical answer

Generate with context

Similar to CorrectiveRAG

Hypothetical Prompt Embeddings (HyPE)



Using rule-based natural language processing to improve disease normalization in biomedical text

Ning Kang, Bharat Singh, Zubair Afzal, Erik M van Mulligen, Jan A Kors

1. Coordination Resolution

I want bigger biceps and triceps

=>

I want bigger biceps and bigger triceps

2. Abbreviation Expansion

I want to improve my VO2 max

=>

I want to improve my maximal oxygen uptake

3. Term Variation Handling

shoulder width exercises

=>

exercises for lateral deltoid width

```
=== Running: Baseline Embedder (no transformations) ===
/Users/hikmetcankoseoglu/Desktop/492/mteb/src/run_benchmark.py:160: DeprecationWarning: MTEB is de
te` function instead.
    evaluation = mteb.MTEB(tasks=[task])
```

Selected tasks

Retrieval
- FitnessRAGRetrieval, *t2t*

=== Results Comparison ===

Metric	Custom	Baseline	Delta
-----	-----	-----	-----
ndcg_at_10	1.0000	1.0000	+0.0000
ndcg_at_5	1.0000	1.0000	+0.0000
recall_at_10	1.0000	1.0000	+0.0000
precision_at_10	0.1000	0.1000	+0.0000

The embedder know the distinction between abbreviations and the full form?

```
"abbreviations": [
  ,
  {
    "term": "BMR",
    "canonical": "Basal Metabolic Rate",
    "mean_similarity": 0.858817,
    "min_similarity": 0.779498,
    "max_similarity": 0.89717,
    "per_template": [
      {
        "template": "What are the benefits of {term}?",
        "informal": "What are the benefits of BMR?",
        "canonical": "What are the benefits of Basal Metabolic Rate?",
        "similarity": 0.779498
      },
      {
        "template": "How does {term} affect performance?",
        "informal": "How does BMR affect performance?",
        "canonical": "How does Basal Metabolic Rate affect performance?",
        "similarity": 0.877393
      },
      {
        "template": "Explain {term} to a beginner.",
        "informal": "Explain BMR to a beginner.",
        "canonical": "Explain Basal Metabolic Rate to a beginner.",
        "similarity": 0.880435
      },
      {
        "template": "What is the best way to improve {term}?",
        "informal": "What is the best way to improve BMR?",
        "canonical": "What is the best way to improve Basal Metabolic Rate?",
        "similarity": 0.89717
      },
      {
        "template": "How is {term} measured?",
        "informal": "How is BMR measured?",
        "canonical": "How is Basal Metabolic Rate measured?",
        "similarity": 0.859589
      }
    ]
  }
],
```

> Find

=== Abbreviations (27 matched, 22 skipped) ===

Term	Canonical	N	Similarity
LISS	Low Intensity Steady State	1	0.9264
FTP	Functional Threshold Power	3	0.9528
BB	Barbell	11	0.9713
MVC	Maximum Voluntary Contraction	1	0.9726
DOMS	Delayed Onset Muscle Soreness	5	0.9733
DB	Dumbbell	6	0.9761
TDEE	Total Daily Energy Expenditure	4	0.9782
HIIT	High Intensity Interval Training	3	0.9796
RPM	Revolutions Per Minute	1	0.9808
SAQ	Speed, Agility, Quickness Training	1	0.9815
MPS	Muscle Protein Synthesis	9	0.9823
RPE	Rate of Perceived Exertion	8	0.9845
TUT	Time Under Tension	6	0.9869
GPP	General Physical Preparedness	1	0.9870
VO2MAX	Maximal Oxygen Uptake	12	0.9898
HRR	Heart Rate Reserve	1	0.9912
ATP	Adenosine Triphosphate	3	0.9942
LBM	Lean Body Mass	1	0.9943
PNF	Proprioceptive Neuromuscular Facilitat	2	0.9945
SAID	Specific Adaptation to Imposed Demands	1	0.9955
1RM	One Repetition Maximum	11	0.9958
CNS	Central Nervous System	8	0.9960
RM	Repetition Maximum	15	0.9962
KG	Kilogram	20	0.9969
OHP	Overhead Press	2	0.9976
KCAL	Kilocalorie	4	0.9983
ROM	Range Of Motion	16	0.9985

No corpus match (skipped): EMOM, AMRAP, MET, HRMAX, BMR, BMI, BCAA, RIR, PR, PB, PCR, EPOC, LB, ECG, EMG, RSI, RMR, HIT, SS, PL, WL, BF

Mean: 0.9842 Min: 0.9264 Max: 0.9985

=== Lay Terms (7 matched, 4 skipped) ===

Term	Canonical	N	Similarity
calves	triceps surae	4	0.9927
pecs	pectoralis major	1	0.9937
abs	rectus abdominis	3	0.9962
quads	quadriceps femoris	5	0.9965
lats	latissimus dorsi	3	0.9974
hammies	hamstrings	5	0.9974
traps	trapezius	5	0.9980

The dataset is too easy, we can add distractors that contains similar vocabulary but doesn't answer query

Add distractors, now there are 400 docs (300 distractors)

```
=== Results Comparison ===
```

Metric	Custom	Baseline	Delta
ndcg_at_10	1.0000	0.9963	+0.0037
ndcg_at_5	1.0000	0.9963	+0.0037
recall_at_10	1.0000	1.0000	+0.0000
precision_at_10	0.1000	0.1000	+0.0000

Next?

RTEB Benchmark (Sports Science)

Reranker?